PV101 Day 4

Small solar system build: panel, controller, 12.8V LiFePO4 battery, wiring, inverter

VERIFY FIRST. CONNECT SECOND.

5 kits. 6 students. One shared station. Every build gets peer-checked before power.

Today the goal is not “make the inverter light up.” The goal is a safe, traceable system.

Today's flow

Quiz -> system pieces -> wiring sequence -> kit build



Morning: make the system understandable before anyone touches a conductor.
Afternoon: build five small systems with one two-student shared kit.
End: every kit is shut down, measured, documented, and returned safely.

The kit parts

Each station builds one small solar system

PV panel source

Charge controller

12.8V LiFePO4 50Ah battery

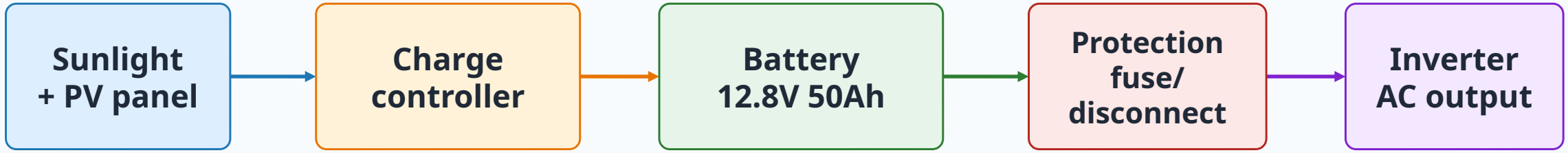
Fuse / disconnect

Inverter DC -> AC

One PV panel feeds the charge controller.
The battery stores energy and stabilizes the system.
The inverter draws from the battery side through protection.
Labels and meter readings make the system serviceable.

System path

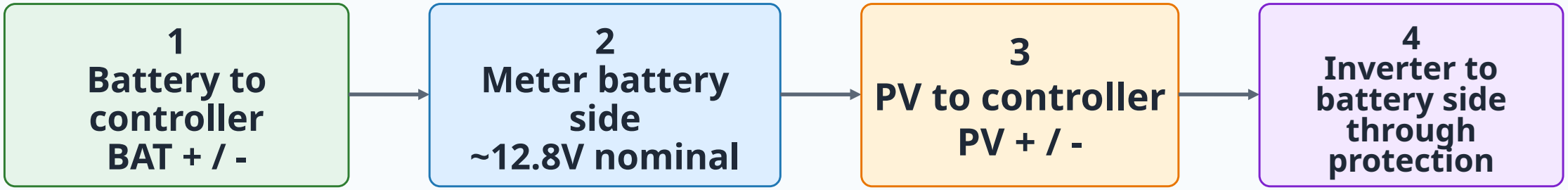
Trace it before building it



Every positive path needs a matching return path. Every power path needs a safe way to stop.

Connection order

Most controllers want battery first, then PV



Keep the PV panel covered or disconnected while wiring.
Keep the battery fuse open/removed until approval.
When unsure, stop. Solar equipment is not impressed by confidence.

Station roles

5 kits for 6 students

Stations 1-4
solo builder

Station 5
2-student shared kit

All stations
peer-check before power

Two-student station swaps Builder and Inspector roles halfway through. Solo builders call over a neighbor for peer-check before energizing. Nobody approves their own mystery wiring. Civilization has suffered enough.

Polarity check

Do not trust colors alone

Battery + / -

Controller BAT + / -

Controller PV + / -

Inverter + / -

Call the meter setup: DC volts, correct lead ports, expected range.
Say the reading and probe placement out loud.
A reversed connection can damage controller or inverter hardware.

Protection matters

A 50Ah battery is not a toy just because it fits on a bench

Fuse close to battery

Correct conductor path

Visible disconnect

No loose copper

Protection exists to protect conductors and equipment from fault current.
Put protection where it protects the most wire.
Open/remove protection before changing wiring.

Inverter rule

The inverter belongs on the battery side through protection

Good path
Battery -> fuse -> inverter

Bad habit
Controller load terminals -> inverter

Inverters can draw high current. Controller load terminals are often for small DC loads, not “whatever fits.”

Mechanical quality

Good electrical work starts as good physical work

Strip without nicking strands.
Use the correct crimp die and complete the ratchet cycle.
Tug test every crimp.
Route wires so terminals are not holding mechanical strain.
Leave enough slack for service, but not a spaghetti shrine.

Bad crimp? Cut it off and redo it.

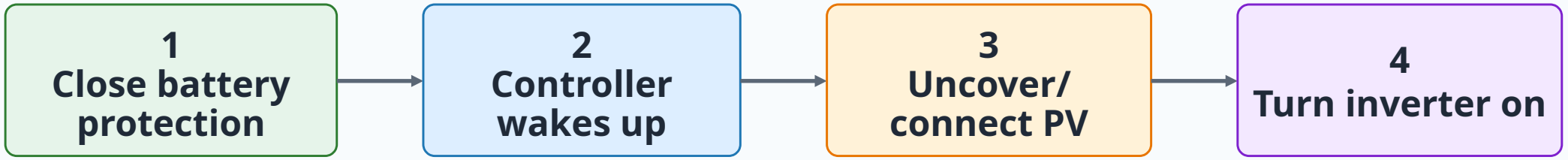
Commissioning checklist

A kit earns power only after the checks

PV covered or disconnected during wiring.
Battery fuse/breaker open until approval.
Battery voltage measured directly and at controller.
PV polarity and voltage measured before final connection.
Inverter input path traced and protected.
Peer check complete, then instructor approval.

Energizing order

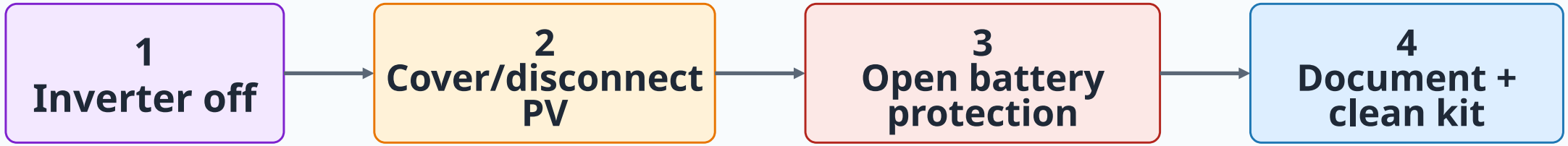
Slow sequence. Clear callouts. No surprises.



Test only with an approved small load. No creative experiments. This is class, not a mythology generator.

Shutdown order

End safe, not merely “done”



Record readings before forgetting them, the classic human speedrun.
Return all kit parts together.
Leave no energized mystery wiring.

Measurement log

What students write down

Battery direct	VDC + polarity	Measured: _____
Controller BAT	VDC + polarity	Measured: _____
PV open circuit	VDC + polarity	Measured: _____
Controller PV	VDC + polarity	Measured: _____
Inverter input	VDC + protection	Measured: _____
AC output	approved load only	Measured: _____

No measurement, no approval. The electrons can wait.

Exit ticket

Three checks before dismissal

1. Trace: PV -> controller -> battery -> inverter -> load.

2. What connects first on most controllers: battery or PV? Why?

3. Name one STOP condition from today's lab.

Tools counted. Kits safe. Panels covered. Batteries protected.